COMP4423 Assignment 2

PolyU Main Campus Image Classification

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Code: https://github.com/Retr0-rgb-lab/COMP4423_Assignment2

Dataset link (the same string as `DATASET_LINK.txt`):
https://drive.google.com/drive/folders/1_ld7gsHLhzyFJY3bUgTIwoOeVXRDFuAr

The complete photograph set is on Google Drive as 27 zip parts named `COMP4423_A2_dataset_part01.zip` through `COMP4423_A2_dataset_part27.zip`. They are one archive cut into 27 byte chunks, not 27 separate archives, so join them before unpacking: `cat COMP4423_A2_dataset_part*.zip > COMP4423_A2_dataset.zip` on Linux or macOS, `copy /b` on Windows, then `unzip COMP4423_A2_dataset.zip`. The file names inside are the ones used in this report. The link is shared as *anyone with the link*, so no sign-in is needed; each part is over 100 MB, so Google shows a *Download anyway* button that requires no account.

0. What the assignment asks, and where this report answers it

Requirement	Where answered
Task 1 — class definitions, collection plan, annotated map	§§1.2, 2.1; Figures 2–3
Task 2 — labelled dataset, location-group split, counts	§§2.1–2.3, 2.5–2.6
Task 3 — features, traditional classifier, training evidence	§3 (tables in 3.1, 3.3, 3.4, 3.6)
Task 4 — test metrics, examples, failure modes	§§4–5
Task 5 — local app, same preprocessing, clear launch	§6
Task 6 — template questions answered	§§7–8

Section 7 answers the six template questions in the template's own wording, as short answers that cross-reference earlier sections.

1. Introduction

1.1 Problem and motivation

The program takes one scene photograph and predicts whether the camera was inside the Hong Kong Polytechnic University main campus. The positive class is that main campus only. A building that belongs to PolyU but sits off the main campus is outside. The label is geographic. It is not "indoors versus outdoors", and it is not "red brick versus something else".

The practical question is whether a traditional classifier, reading pixels alone, can tell those two geographies apart when many scenes look alike: walkways, courtyards, lawns, and red-brick façades exist on both sides of the boundary. The Hong Kong University Main Building was collected for that reason. It is red brick, and it is outside PolyU.

1.2 Class definitions

The two classes are the ones the official campus map defines, with no local reinterpretation.

Label	Definition used in this project
inside (1)	The camera and the main scene are inside the PolyU main campus, using the official campus map as the boundary [10].
outside (0)	The camera and the main scene are outside that campus. Affiliated sites off the main campus stay outside.

Uncertain boundary views are excluded: gate thresholds, covered turnstiles at the entrance, connecting bridges, and shots that look across the boundary. The camera and the subject must

be on the same side. Labels are not assigned from red brick, a logo, or architectural style. The classifier receives pixels only. GPS, EXIF, file names, folder names, and map lookups may be used to check a label or to find a file. They are not features.

At the main entrance, 11 photographs were removed under this rule: 4 taken at the campus name board, where the side of the gate is ambiguous, and 7 in the covered passage at the turnstiles, which is a threshold and also contains recognizable people.

1.3 Pipeline in one pass

1. Photographs are grouped by physical place. One place is one `group_id`.
2. Groups are split into train, validation, and test. A group is never split.
3. Each training and evaluation image is resized and center-cropped to 256×256 (Section 3.2).
4. HOG is computed on the grayscale crop, with the block normalisation the descriptor applies internally. The random forest needs no further scaling, so the locked pipeline has no scaler; the linear SVM and logistic-regression rows do have one, fitted inside each fold. The locked model uses this descriptor alone; HOG+LBP, LBP alone, a colour histogram and two bag-of-words variants were also measured and are the comparison features of Section 4.1.
5. A random forest (300 trees, maximum depth 12) outputs outside or inside.
6. The fitted object is `models/task3_pipeline.joblib`.

Figure 1 shows the same path as a diagram, together with the order in which the model was chosen.

Two things about the figure are worth stating in words. The left-hand column is a single path: the photograph goes through one resizer, one feature extractor and one classifier, and the object that comes out is the file the application loads. There is no second path: the application loads the same saved pipeline and applies the same preprocessing as evaluation. The right-hand column is ordered: the feature comparison and the classifier comparison both run on the training places alone, in two stages, and only after both have finished is the test set read, once. The red line in the figure marks that boundary, and everything about how the result in Section 4 should be read follows from it.

Deep networks and CNN features are not used. The classifier reads pixels only.

1.4 Local application

Section 6 is the desktop program that loads `models/task3_pipeline.joblib` and shows one photograph's prediction. Sections 4 and 5 are the validation report and the single test evaluation of that same object. The application does not choose features and does not rescore the test set.

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2. Dataset

2.1 Places and the collection strategy

One physical place is one location group. Nearby views of the same courtyard, building, or landmark stay together. Separate buildings in the same district do not. HKU's modern teaching blocks and the Main Building are different groups. West Kowloon's M+ museum, Art Park, and harbourfront promenade are different groups. Continuous PolyU walkways stay in one group.

Inside the group, variation is recorded in columns rather than new groups: `sub_area` for the part of the place, and `scene_class` for one of 13 scene types. Shooting variation also stays

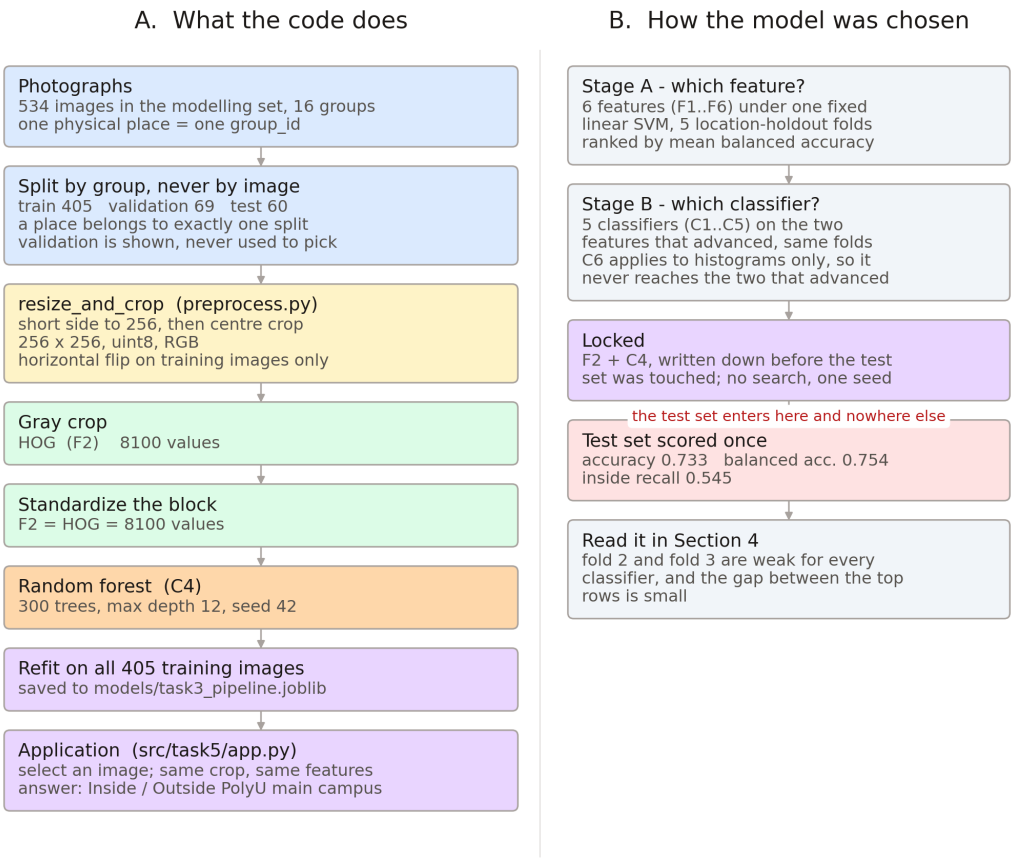


Figure 1. The pipeline, and the order in which the model was chosen. Left: what the code does, from the photographs through the split, the crop, the feature and the random forest to the saved pipeline that the application loads. Right: how the model was chosen. The comparison runs on the training places alone, in two stages, and the test set is reached once, after the choice has been written down. Six features were compared in Stage A and five classifiers in Stage B; the diagram names the winner, not every candidate.

inside the group — making any axis its own group would turn the split into a split on that axis. Night frames stay in the daytime group of the same place. What was actually varied during collection:

Axis	How it was varied
Viewpoint	Frontal views, oblique side angles, and different camera heights.
Distance	Near, medium, and far stand-off from the same place.
Lighting	Daytime and night frames; night stays in the same location group as the daytime shots of that place.
Background	Buildings, walkways, and green space were collected on both classes so scene type would not become a class cue.

Collection quotas were within-group targets; final counts are below and in Section 2.5.

Both classes contain buildings, covered walkways, green space, and open paving. Eleven of the thirteen scene types appear on both sides of the label. The two types that appear only outside are deliberate: five indoor frames of an HKU corridor (so "indoors" cannot become a synonym for inside), and nine market-stall frames. Among the 716 usable photographs before the per-group cap, façades and covered walkways dominate (149 and 128); road frontage is unbalanced (10 inside, 57 outside); water has 19; construction hoarding is temporary. Night frames are 31/304 usable inside (10.2%) and 56/412 usable outside (13.6%). In the modelling set, all thirty outside night frames are in training (OUT-01, OUT-03); the seven inside night frames in test are all from IN-05. Limits are discussed in Section 2.6. The full per-place kept counts and splits are in the main table of Section 2.5.

Figure 2 puts the seven inside places on the campus map; the table under it is the inside half of that main place list.



Figure 2. The seven inside places on a PolyU campus map, one numbered pin per place. The pin numbers are the row numbers in the table below. Marker 05 is on the south side of J Core; the place it marks is the main-entrance forecourt with its inner lawn and clock tower. The 11 ambiguous gate frames are in the excluded set; the 16 frames that remain are on the campus side of the boundary.

No.	Group	Place	Split	Usable images
01	IN-01	Walkway, J block to VA block	train	72
02	IN-02	Coffered walkway, P/Q block	train	62
03	IN-03	University Library, University Gallery and the E block	train	44
04	IN-04	FSCH, VS block and swimming pool	train	53
05	IN-05	Main entrance forecourt and clock tower	test	16
06	IN-06	Building courtyard and Block A entrance	val	40
07	IN-07	CMA courtyard	test	17
		total, before the per-group cap		**304**

The counts are usable images with status include; they sum to 304 before the 50-image cap and class balancing. Outside places are spread over three districts, so Figure 3 places all sixteen groups on one map (nine outside on a city map; seven inside on a campus inset). Filled circle = training, ring = validation, square = test; colour is the class. Per-place kept counts for every outside group are in Section 2.5.

2.2 Hard negatives

Red is the easiest shortcut. Much of the main campus is red brick, and a color histogram can score well by voting "red means inside". Outside photographs were chosen to block specific inside appearances:

Outside group	Inside appearance it matches
OUT-06 HKU Main Building (test)	Red-brick façades at IN-01, IN-03, and IN-04. Edwardian red brick and granite. Paired with IN-05, the main entrance forecourt, which is also in the test set.

Outside group	Inside appearance it matches
OUT-02 HKU modern blocks and links (train)	Coffered walkway IN-02 and covered links IN-06. Includes the five indoor corridors.
OUT-09 West Kowloon harbourfront (val)	Entrance steps IN-05 and the large stair at IN-06.
OUT-08 Art Park (train)	Open lawns and courts at IN-02, IN-06, and IN-07. Bound to the same split as OUT-05 because the two sites see each other across a road.
OUT-05 M+ (train)	Modern buildings and hard plazas.
OUT-04 Elements (train)	Large roofs over open space, compared with the coffered ceiling at IN-02.
OUT-03 Kowloon Park (train)	Lawns, courtyards, and water. Eleven red-tile Chinese-garden frames are an extra red-brick negative.
OUT-07 HKU campus green (test)	Lawns at IN-05 and the courtyard green at IN-07. Not a red-brick negative. Paired with IN-07 in the test set.
OUT-01 Jordan (train)	Dense street façades, as a contrast to campus blocks.

2.3 Labels, exclusions, and anchors

734 photographs were annotated. 18 were excluded. 716 remained, in 16 places (7 inside, 9 outside). After a cap of 50 images per group and class balancing, the modelling set has 534 images, 267 inside and 267 outside. Section 2.7 records why the count changed on 2026-10-27.

Excluded	Count	Reason
IN-05	11	Name board at the gate (4) or covered turnstile passage (7). Notes on the manifest rows match this split.
IN-01	2	The subject is the Communal Building, which no location group covers, and the Library block is in the background; the place could not be established from the frame (Section 2.7).
OUT-01	2	A readable license plate in the foreground. Both frames are excluded on privacy grounds whatever the plate says; the readings are kept on the manifest rows.

Excluded	Count	Reason
OUT-02	2	Street signs on the outer side of the campus boundary.
OUT-03	1	Several recognizable faces.

Exclusion was decided on the photograph, and never on the strength of a model score.

Phone transfer stripped EXIF GPS. Ground truth anchors are signs that are visible in the frame, including the university library sign and crest (IN-03 `IMG_1459`), Stanley Ho Building (IN-04 `IMG_1229`), CMA Building and Lui Che Woo Building (IN-07 `IMG_1428`), "THE UNIVERSITY OF HONG KONG" on the Main Building (OUT-06 `IMG_0863`), and Jordan Road (OUT-01 `IMG_1002`). Every anchor is a sign that is visible in the frame, so each label can be re-checked from the photograph alone. The campus name board `IMG_1304` is excluded and is not used as an anchor.

2.4 Why this scale is enough, and where it is not

There is no required image count. The unit that has to be learned and tested is the place, not the frame. Sixteen places are enough to put several places of each class in training, hold out paired places for a five-fold location check, and still leave a validation pair and a test pair. Adding more frames inside a place that is already in training does not add an unseen place.

The cap and the balance step exist so that one large place cannot dominate the pixel count. OUT-01 entered with 82 usable frames and is stored as 50. The smallest place, OUT-07, has only 7 frames, which also makes the image counts uneven: the test outside side holds 27 images against 33 inside images. Sixteen places are enough to run the protocol. They are not enough to make the test estimate sharp: the test set can hold only four places.

2.5 Split, leakage controls, and counts

Compliance with the assignment split is judged by **location-group shares aiming at ≈60/20/20**, with both classes in every split. Image proportions (about 75.8% / 12.9% / 11.2%) are a by-product of unequal group sizes and the 50-image cap; they are not the target.

`data/annotations/splits.csv` is the split. It is derived from the annotation manifests: each group is capped, the two classes are balanced, and groups are assigned to splits. Automatic checks verify that each location group belongs to exactly one split, that OUT-05 and OUT-08 stay together, and that every split contains both classes. Augmentation is horizontal flip, applied only to training images after the split. Subsampling inside a capped group is evenly spaced so the viewpoint range remains.

The split below was re-derived on 2026-10-27, after a photo-level re-check found three grouping errors across four groups (Section 2.7). The rule gives training the largest groups and hands the rest between validation and test; the two mutually visible West Kowloon groups are one unit. The repair changed the split of three groups and the stored size of five, so the assignment moved with those sizes.

Image counts and groups, from `data/annotations/dataset_summary.md`:

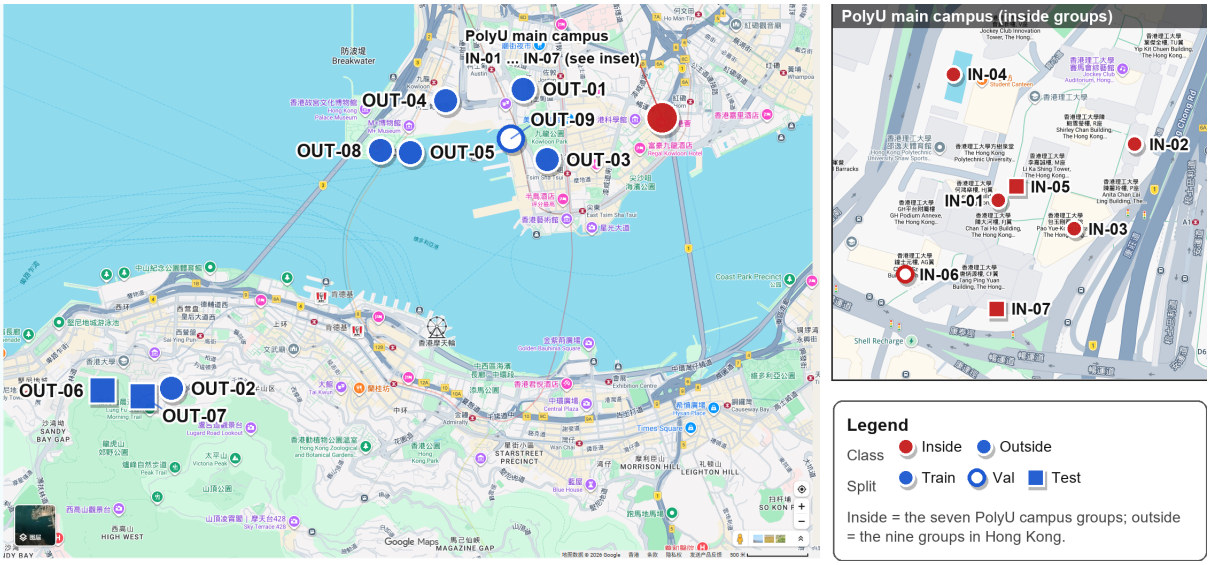


Figure 3. The sixteen collection locations. Left, Hong Kong, with the nine outside groups. Right, a campus inset with the seven inside groups. Colour is the class and shape is the split. The base map is a screenshot of Google Maps; every pin is placed from a named feature of the map for that place.

Split	Class	Images	Share of the 534	Groups	Share of the 16	Group ids
train	outside	211	39.5%	6	37.5%	OUT-01, OUT-02, OUT-03, OUT-04, OUT-05, OUT-08
train	inside	194	36.3%	4	25.0%	IN-01, IN-02, IN-03, IN-04
val	outside	29	5.4%	1	6.3%	OUT-09
val	inside	40	7.5%	1	6.3%	IN-06
test	outside	27	5.1%	2	12.5%	OUT-06, OUT-07
test	inside	33	6.2%	2	12.5%	IN-05, IN-07
all	outside	**267**	**50.0%**	**9**	**56.3%**	OUT-01 … OUT-09
all	inside	**267**	**50.0%**	**7**	**43.8%**	IN-01 … IN-07

Totals: train 405, validation 69, test 60, sum 534. By groups: 10/16, 2/16, 4/16 (62.5%, 12.5%, 25%). With 16 indivisible groups, 20% is 3.2, so the split lands at 10, 2 and 4. Validation is the

side below target because holding out one group per class for validation, after two per class for test, was preferred over taking a second group out of training on each side; the validation score therefore rests on one place per class and is reported, not used, as the selection score. The test outside side is two groups because each is paired with a resembling test inside place (Section 2.2).

Main place table (usable → kept after the per-group cap). Inside rows match Figure 2; outside rows complete the sixteen groups.

Group	Place	Class	Usable	Kept	Split
IN-01	Walkway, J Core to VA	inside	72	50	train
IN-02	P/Q coffered walkway	inside	62	50	train
IN-03	University Library, Gallery and E block	inside	44	44	train
IN-04	FSCH / VS / pool	inside	53	50	train
IN-05	Main entrance forecourt and clock tower	inside	16	16	test
IN-06	Block court and Core A entrance	inside	40	40	val
IN-07	CMA Building courtyard	inside	17	17	test
OUT-01	Jordan	outside	82	50	train
OUT-02	HKU modern blocks and links	outside	57	50	train
OUT-03	Kowloon Park	outside	72	32	train
OUT-04	Elements	outside	62	29	train
OUT-05	West Kowloon, M+	outside	21	21	train
OUT-06	HKU Main Building	outside	20	20	test

Group	Place	Class	Usable	Kept	Split
OUT-07	HKU campus green	outside	7	7	test
OUT-08	West Kowloon, Art Park	outside	41	29	train
OUT-09	West Kowloon harbourfront	outside	50	29	val

Place names here are the full names. The generated summary in `data/annotations/dataset_summary.md` uses shorter labels for a few outside rows; the group id, class, counts, and split are the same.

2.6 Dataset limits that affect later numbers

- The test estimate rests on four places. A single difficult building moves the accuracy.
- OUT-07 contributes only 7 test images.
- Road frontage is 10 inside versus 57 outside in the usable pool. A model could treat heavy traffic as outside. The four test places are a forecourt, a courtyard, a brick building, and a lawn, so Section 5 does not attribute the test misses to that imbalance.
- Indoor evidence exists only on the outside class, and only as 5 frames.
- Construction hoarding will disappear. It is a durability limit, not a stable campus cue.
- Night shares are 10.2% inside and 13.6% outside of the usable pool, within a few points of each other.

2.7 A location group that was not one place

The group labels were assigned from the collection folders. A photo-level re-check on 2026-10-27 found three errors across four groups:

- **IN-05** mixed three places (walkway frames that belong to IN-01, Library-complex frames that belong to IN-03, and its own main-entrance frames).
- **IN-01** held six frames of the Library / Shirley Chan / Chow Yei Ching block.
- **OUT-08** and **OUT-09** both photographed the FREESPACE building at Art Park from different splits.

That is leakage in the assignment's sense: the same scene on both sides of a split. The repair moved 31 frames to the correct groups and excluded 2 whose place could not be established from the photograph; the split was then re-derived and the experiments were re-run. The per-group before/after sizes and the re-balance arithmetic are in the Appendix.

Consequences for Section 4: the group counts and the test set are those of the repaired split (267/267 after re-cap and re-balance). The Library complex is now a training group; test inside places are the main entrance and the CMA courtyard. The assignment moved with the repaired sizes under the project's own rule (largest groups to train), not by hand after seeing a score. As noted in Section 3.6, this is a corrected evaluation of the collection, not a wholly independent blind test.

3. Method

3.1 Design

The comparison was fixed before the run: the six features, the classifiers, the five folds, the single selection score, and the tie-break rules are locked in the shipped code. A fingerprint of that source and of `splits.csv` is written beside the result tables; a table whose fingerprint does not match is refused, so a result cannot silently come from other code.

Stage A asks which representation helps. Six hand-crafted features share one classifier: standardization, where the classifier needs it, and a linear SVM with `C = 1`. Stage B asks whether the classifier matters. The two features with the highest location-holdout score are each trained with the classifiers in Section 3.4. Parameters are fixed. There is no search on the validation set and none on the test set.

The selection score of a configuration is the mean balanced accuracy of five location-holdout folds. Balanced accuracy is the average of the outside recall and the inside recall, so a fold that only recognizes one class does not look successful. Validation accuracy is written down and is not part of the sort. Configurations are ranked by mean location-holdout balanced accuracy; that ranking was checked against the published tables.

Ties on features keep the smaller dimension, then the earlier id. Ties on classifiers keep the simpler one, in the order logistic regression, linear SVM, naïve Bayes, k-NN, random forest, gradient boosting. One planned row was conditional on the winner being a linear SVM — a 100-component PCA followed by an RBF SVM, fit inside the fold — and it was not run, because the winner was not.

The table sets out the choices that decided the pipeline, the alternatives that were considered, and the evidence each answer rests on. Where the project cannot support an answer, the cell says so instead of asserting one.

Decision	Alternatives considered	Why this one	Evidence
The split unit is the location group	random per-image split; split by scene class	The model must work on a place it has not seen, so a place is what gets held out.	Automatic checks verify each group belongs to exactly one split; Section 2.5
Traditional ML only, no neural network	pretrained CNN features; fine-tuning	The assignment excludes deep learning for Task 3.	Assignment requirements

Decision	Alternatives considered	Why this one	Evidence
256×256 centre crop	stretch to 256×256; multi-scale; 224 or 320 px	HOG's cells and blocks fix the crop geometry, and one resize-and-crop path serves training, evaluation and the app, so the three cannot disagree. Cost: a portrait frame loses strips top and bottom.	Section 3.2 (input size and crop); Section 6.3
HOG on the grayscale crop	HOG+LBP; LBP; colour; bag of words; BoW + pyramid	Shape beat the rest on the holdout: F2 0.701 against F3 0.607, F6 0.544, F1 0.537, F5 0.531. Adding LBP moved it +0.007 (SVM) and −0.009 (forest), both inside the fold spread.	Section 4.1
Rank by location-holdout balanced accuracy	validation accuracy; test accuracy; plain accuracy	The score must be measured on unseen places, and balanced accuracy stops a one-class answer from looking good.	Section 4.1
Two stages: features, then classifiers	the full 6×6 grid; random search	Stage A moves the representation with the classifier fixed, Stage B the reverse. The representation moves the score more than the classifier does.	Sections 4.1, 4.2
No hyperparameter search	grid over `C`, depth, trees, HOG cell, LBP radius, vocabulary	Fixed before the run to keep the protocol pre-registered and single-seed. The F2 fold spread, 0.471 to 0.860, dwarfs the 0.009 between the top rows, so no row is claimed optimal and tuning was not measured.	Section 4.2

Decision	Alternatives considered	Why this one	Evidence
Horizontal flip, training only, after the split	colour jitter; rotation; scale; none	An augmentation that moves the camera would blur the place boundary under test; colour jitter would hide the lighting axis that was collected.	Section 2.1

3.2 Preprocessing

Every training, evaluation, and application image goes through one resize-and-crop path:

1. Open the file as RGB with Pillow. OpenCV's BGR reader is not used, so a later interface cannot silently disagree with evaluation.
2. Scale the short side to 256 with area interpolation, keeping the aspect ratio.
3. Center-crop to 256×256. A landscape frame loses equal strips on the left and right. A portrait frame loses equal strips on the top and bottom.
4. Return `uint8` of shape 256×256×3. HOG, LBP, and SIFT convert that crop to gray. The color histogram uses HSV of the same crop.

Horizontal flip is a training augmentation only. Validation and test images are not flipped. Color jitter is not used. Night and lighting were collected on purpose; jitter would hide that axis.

The random seed is 42.

3.3 Features

Id	Feature	Setting	Dimension
F1	HSV histogram	H 16 × S 16 × V 8. Whole frame plus a 2×2 grid. Each histogram L1-normalized, then concatenated.	10240
F2	HOG	9 orientations, 16×16 cells, 2×2 blocks, L2-Hys, on gray [1].	8100
F3	LBP	Uniform pattern, P = 8, R = 1, 10 bins [2]. Whole frame plus 2×2, each block L1-normalized.	50

Id	Feature	Setting	Dimension
F4	HOG + LBP	F2 and F3 standardized separately on the training images, then concatenated.	8150
F5	Bag of words	SIFT [4] keypoints, at most 500, in a bag-of-words histogram [5]. Mini-batch k-means, 256 words, fit on that fold's training descriptors only. Word counts L1-normalized. An image with no keypoints becomes a zero histogram and is counted in the log.	256
F6	Bag of words + spatial pyramid	F5 on the whole frame, plus four histograms on a 2×2 grid [6].	1280

SIFT is `cv2.SIFT_create` from `opencv-contrib` [9]. If that function is missing, the run stops. ORB is not a substitute. In the environment used for the 2026-10-27 run, SIFT was available.

The vocabulary is not stored across folds. A fold that is building a codebook never sees the held-out place.

3.4 Classifiers

Id	Classifier	Setting
C1	Linear SVM	`LinearSVC`, `C = 1`, standardized features
C2	Logistic regression	`C = 1`, `lbfgs`, standardized features
C3	k-NN	k = 5. Cosine distance on histogram features. Euclidean distance when the feature includes HOG.
C4	Random forest	300 trees, `max_depth = 12`, `random_state = 42` [3]

Id	Classifier	Setting
C5	Histogram gradient boosting	`max_depth = 6`, `random_state = 42`
C6	Multinomial naïve Bayes	`alpha = 1`, only for non-negative L1 histograms (F1, F3, F5, F6)

C1, C4, and C5 are the three main families. C2 and C3 are different decision rules. C6 is a simple histogram control; F2 and F4 skip it. C6 was run as a supplementary comparison on the four L1 histograms; its rows do not enter the feature ranking (C1 column only).

Feature	C1 linear SVM	C6 naive Bayes	C6 inside recall	C6 outside recall
F1 colour histogram	0.537	0.532	0.224	0.839
F3 LBP	0.607	0.500	0.000	1.000
F5 bag of words	0.531	0.518	0.048	0.988
F6 bag of words + spatial pyramid	0.544	0.576	0.419	0.733

On F3 and F5, naïve Bayes collapses toward outside (inside recall 0.000 / 0.048). On F1 and F6 it does not (0.224 / 0.419). Every C6 row stays below 0.607, while linear SVM reaches 0.708 and 0.701 on F4 and F2, so the two features that advance are unchanged under both protocols. The order of the remaining features does change; that is why the C6 pass is reported and not used for selection.

3.5 Location holdout

Each training place is a single class, so a fold always holds out one inside place and one outside place together. OUT-05 and OUT-08 are one atom. The five folds are:

Fold	Held-out inside	Held-out outside
0	IN-01	OUT-01
1	IN-02	OUT-02
2	IN-03	OUT-03
3	IN-04	OUT-04
4	IN-01	OUT-05 and OUT-08

IN-01 is held out twice because there are five outside atoms and four inside training groups. The five balanced accuracies are averaged as they are. The two IN-01 folds are not merged. The folds follow the training places in group order, so they are re-derived whenever the split is re-derived; the fold set above is the one that produced the numbers in Section 4.

On each fold the feature model (codebook, standardization, and PCA if any) is fit on the other training images, including their horizontal flips. Flips are not placed in the held-out set. The held-out places are scored with balanced accuracy and with both recalls.

3.6 Locked pipeline and how to rerun it

After Stage B, the winning pair is refit on all training images plus horizontal flips. It is saved with joblib, and the validation split is scored once for the report.

The test set is scored once on the repaired split, and the per-image predictions of that single evaluation are what Section 4.4 and Section 5 are computed from. The fit was chosen on the location holdout and never on the test set, so the selection rule was not informed by these 60 predictions. What limits the reading is that the test places are not all untouched: the repair of Section 2.7 was found by re-reading frames grouped by physical place, and their per-place rates had been computed before that repair. The numbers in Section 4 are therefore a corrected evaluation of the collection, not a wholly independent blind test.

Commands, from the repository root. The environment is Python 3.12 with the pins in `requirements.txt`.

```
python -m src.task3.run_stage_a
python -m src.task3.run_stage_b
python -m src.task3.evaluate
```

A supplementary naive Bayes pass on the four histogram features produced the C6 columns of the table in Section 3.4. Those rows are shipped in `outputs/task3/stage_a_nb.csv`. The pass is not part of the selection: the feature ranking is computed from the linear SVM column of Stage A alone. The helper that produced that table is not included in the code archive.

Pinned packages in `requirements.txt`:

```
numpy==2.4.3
Pillow==12.1.1
opencv-contrib-python==5.0.0.93
scikit-image==0.26.0
scikit-learn==1.8.0
joblib==1.5.3
```

`opencv-contrib-python` is required. The non-contrib wheel has no SIFT, and the feature module exits rather than switching descriptors.

What the saved pipeline actually contains, read back from `models/task3_pipeline.joblib`:

Item	Value
Training images before augmentation	405 (211 outside, 194 inside) in 10 location groups
Rows the forest was fitted on	810: the 405 training images and their 405 horizontal flips
Forest	300 trees, `max_depth = 12`, `max_features = "sqrt"`, `random_state = 42`, `n_jobs = -1`

Item	Value
Feature vector fed to the forest	8100 dimensions: the HOG block as computed, with no separate scaler. The saved pipeline has two steps, `features` then `clf`
Saved object	`models/task3_pipeline.joblib`, 2 936 221 bytes (2.80 MiB), two steps: `features` then `clf`
Inference cost	0.68 s per photograph, measured on 20 test frames on the machine used: file read, crop, HOG, `predict` and `predict_proba` (Section 6.3)

The row count follows from the data: the training split holds 405 images, augmentation doubles it, and the fitted forest reports 810 samples. Validation and test images are never flipped, so their 69 and 60 images are scored as they are.

The three pipeline commands were run on 2026-10-27, on the repaired split of Section 2.7, from the code in the archive, together with a local supplementary naive Bayes pass whose table is shipped under `outputs/task3/`; the numbers in Section 4 are the ones that run produced. The stamp beside the results matches the shipped source, so a rerun resumes instead of prompting for `--force`. Every number in Section 4 is printed in full here as well.

The code archive ships the training and evaluation pipeline, the desktop application, the split and manifests, the locked model, the result tables, and the report figures. Helper and plotting scripts used while preparing the report are not included. Implementation map (where to look; the methods themselves are in Sections 3.2–3.5):

Role	Location
Stage A / B training and evaluation commands	Section 3.6 commands above; `src/task3/`
Supplementary naïve Bayes results	`outputs/task3/stage_a_nb.csv` (helper script not shipped)
Saved locked pipeline	`models/task3_pipeline.joblib`
Split file	`data/annotations/splits.csv`
Dataset construction / location repair	done offline; Section 2.7 describes the repair (scripts not shipped)
Result figures (confusion matrices, per-group bars, example grid)	`docs/figures/` (plotting scripts not shipped)
Desktop application	`python -m src.task5.app` (Section 6)

The application loads the saved pipeline and applies the same preprocessing as evaluation; it does not fit a new model.

4. Experimental results

Selection uses location-holdout balanced accuracy only. The validation column is display. The figures in this section are drawn from the locked counts, and every number in the tables below is printed in full in this section.

The comparison is a set of configurations rather than an exhaustive search. Stage one fixes the classifier at the linear SVM so that the feature axis is isolated; stage two varies the classifier only on the two features that advanced. Every classifier was run at one fixed setting, with no parameter search anywhere in the protocol.

4.1 Stage A: linear SVM, six features

Feature	Location-holdout BA	Inside recall	Outside recall	Validation accuracy
F4 HOG + LBP	0.708	0.676	0.739	0.841
F2 HOG	0.701	0.665	0.737	0.797
F3 LBP	0.607	0.544	0.671	0.826
F6 bag of words + pyramid	0.544	0.452	0.635	0.681
F1 color histogram	0.537	0.511	0.564	0.725
F5 bag of words	0.531	0.461	0.601	0.667

F4 and F2 enter Stage B.

The per-fold balanced accuracies behind those means are the five fold columns of the table below, folded as in Section 3.5.

Feature	fold 0	fold 1	fold 2	fold 3	fold 4	mean
F1 HSV histogram	0.600	0.350	0.572	0.704	0.460	0.537
F2 HOG	0.790	0.760	0.527	0.587	0.840	0.701
F3 LBP	0.670	0.470	0.705	0.563	0.630	0.607
F4 HOG + LBP	0.810	0.750	0.531	0.607	0.840	0.708
F5 bag of words	0.580	0.600	0.527	0.558	0.390	0.531
F6 bag of words + pyramid	0.610	0.610	0.499	0.529	0.470	0.544

The color histogram is near the bottom, 0.537 against 0.708 for the winner. That matches the reason HKU was collected: red brick is on both sides of the label, and color does not survive a change of place. Bag of words is last, 0.531, and the spatial pyramid lifts it to 0.544, so the finer

spatial layout buys 0.013, a small gain and still far below the two features that advanced. The validation column and the holdout column do not pick the same pair: validation puts F3, the LBP texture, second at 0.826, while the holdout puts F2, the HOG shape, second at 0.701. Had validation been the selection score, F3 would have gone to Stage B and F2 would not. Validation accuracy is the wrong column to maximize here, and it is shown so that the disagreement stays visible.

4.2 Stage B: F4 and F2 across classifiers

C6 does not apply to F2 or F4. The winner is not a linear SVM, so the RBF row was not run.

Feature	Clf	Location-holdout BA	Inside recall	Outside recall	Validation accuracy
F2	C4 random forest	**0.722**	0.622	0.821	0.783
F4	C4 random forest	0.712	0.627	0.797	0.841
F4	C2 logistic regression	0.712	0.660	0.764	0.826
F4	C1 linear SVM	0.708	0.676	0.739	0.841
F2	C2 logistic regression	0.706	0.660	0.751	0.797
F4	C3 k-NN	0.703	0.562	0.845	0.826
F2	C1 linear SVM	0.701	0.665	0.737	0.797
F2	C3 k-NN	0.689	0.527	0.851	0.754
F4	C5 histogram gradient boosting	0.679	0.606	0.752	0.899
F2	C5 histogram gradient boosting	0.674	0.593	0.756	0.812

The locked pipeline is **F2 HOG + C4 random forest**.

On F2, the forest's validation accuracy (0.783) is below the linear SVM's (0.797); the forest wins because location-holdout BA is higher (0.722 against 0.701). k-NN's outside recall is high (0.851) with inside recall 0.527; on fold 3 it scores 0.471 while the forest scores 0.638.

Fold tables (folds of Section 3.5):

F4 classifier	fold 0	fold 1	fold 2	fold 3	fold 4	mean
C1 linear SVM	0.810	0.750	0.531	0.607	0.840	0.708
C2 logistic regression	0.810	0.760	0.562	0.587	0.840	0.712
C3 k-NN, k=5	0.800	0.730	0.578	0.498	0.910	0.703

F4 classifier	fold 0	fold 1	fold 2	fold 3	fold 4	mean
C4 random forest	0.740	0.740	**0.567**	0.646	0.870	0.712
C5 histogram gradient boosting	0.750	0.730	0.528	0.567	0.820	0.679

Feature	Clf	fold 0	fold 1	fold 2	fold 3	fold 4	mean
F2	C1 linear SVM	0.790	0.760	0.527	0.587	0.840	0.701
F2	C2 logistic regression	0.780	0.790	0.531	0.577	0.850	0.706
F2	C3 k-NN, k=5	0.800	0.770	0.574	0.471	0.830	0.689
F2	C4 random forest	0.750	0.800	0.560	0.638	0.860	0.722
F2	C5 histogram gradient boosting	0.740	0.760	0.479	0.584	0.810	0.674

On F4 the forest does not win every fold, but it has the highest floor (worst fold 0.567); selection ranks by mean. On F2 the forest wins at 0.722. F4 forest vs F4 logistic both print as 0.712 (unrounded 0.712456 vs 0.711810); neither reaches F2+forest. Fold-to-fold spread (e.g. selected config 0.560–0.860) dwarfs the 0.009 margin that locked F2 over F4 with the forest. One seed (42); IN-01 is held out in two folds. Representation moves the score more than the classifier: colour→HOG under linear SVM is +0.164; within F2 the classifier span is 0.047.

4.3 Validation after the refit

The forest was refit on the full training split, with horizontal flips, and then applied to the 69 validation images (IN-06 40, OUT-09 29).

Accuracy **0.783**. Balanced accuracy **0.784**.

	Predicted outside	Predicted inside
True outside	23	6
True inside	9	31

Class	Precision	Recall	F1	Support
outside	0.719	0.793	0.754	29
inside	0.838	0.775	0.805	40

These three-decimal figures are the published rounding of the matrix: 54/69 = 0.783, outside recall 23/29 rounds to 0.793, inside recall 31/40 rounds to 0.775. Nine campus frames are called outside and six harbourfront frames are called inside. Figure 4 draws the same four cells, and each cell also carries its share of the true-class row. Section 2.5 is why 0.783 is not quoted below as generalization.

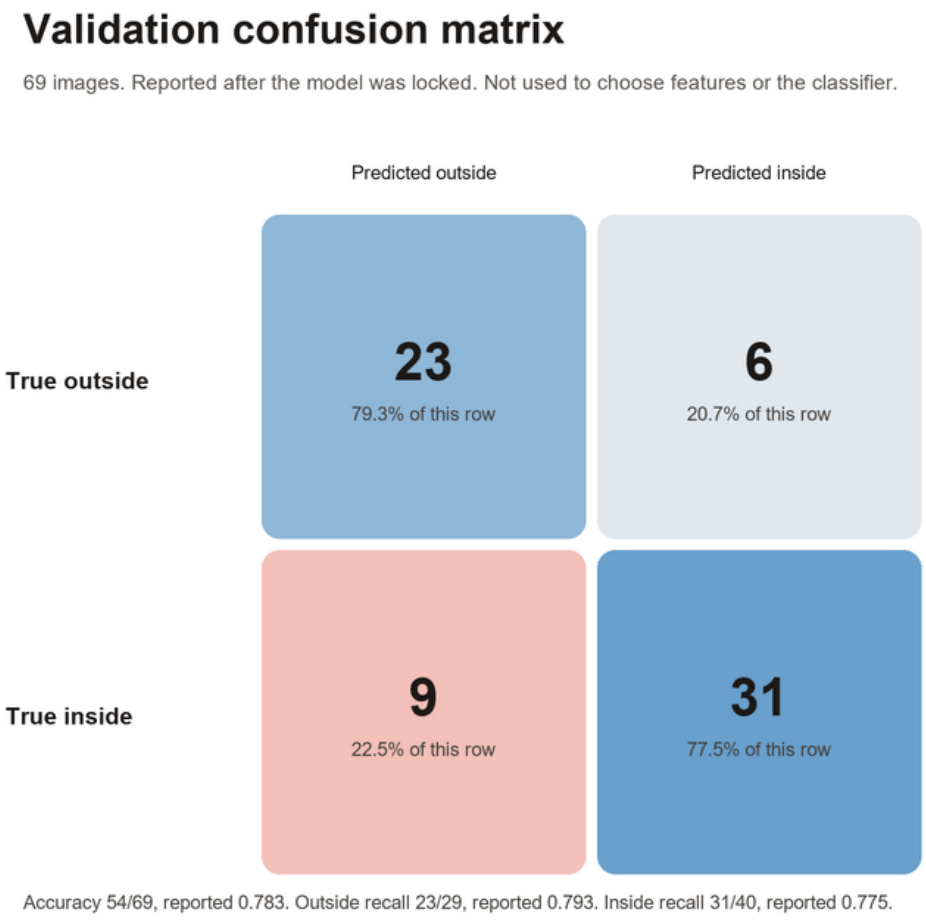


Figure 4. Validation confusion matrix of the locked pipeline. The four counts are the ones in the table above. Not used to choose the model.

4.4 Test, one evaluation on the repaired split

60 images: IN-05 16, IN-07 17, OUT-06 20, OUT-07 7.

Accuracy **0.733**. Balanced accuracy **0.754**.

	Predicted outside	Predicted inside
True outside	26	1
True inside	15	18

Class	Precision	Recall	F1	Support
outside	0.634	0.963	0.765	27
inside	0.947	0.545	0.692	33

Working from the same cells: accuracy 44/60 rounds to 0.733 and balanced accuracy to 0.754; the recalls are 26/27 = 0.963 outside and 18/33 = 0.545 inside, with precisions 26/41 = 0.634 and 18/19 = 0.947. Figure 5 draws the same four cells the same way. The same matrix recomputes from the per-image predictions of that evaluation (outside 0.6341 / 0.9630 / 0.7647, inside 0.9474 / 0.5455 / 0.6923).

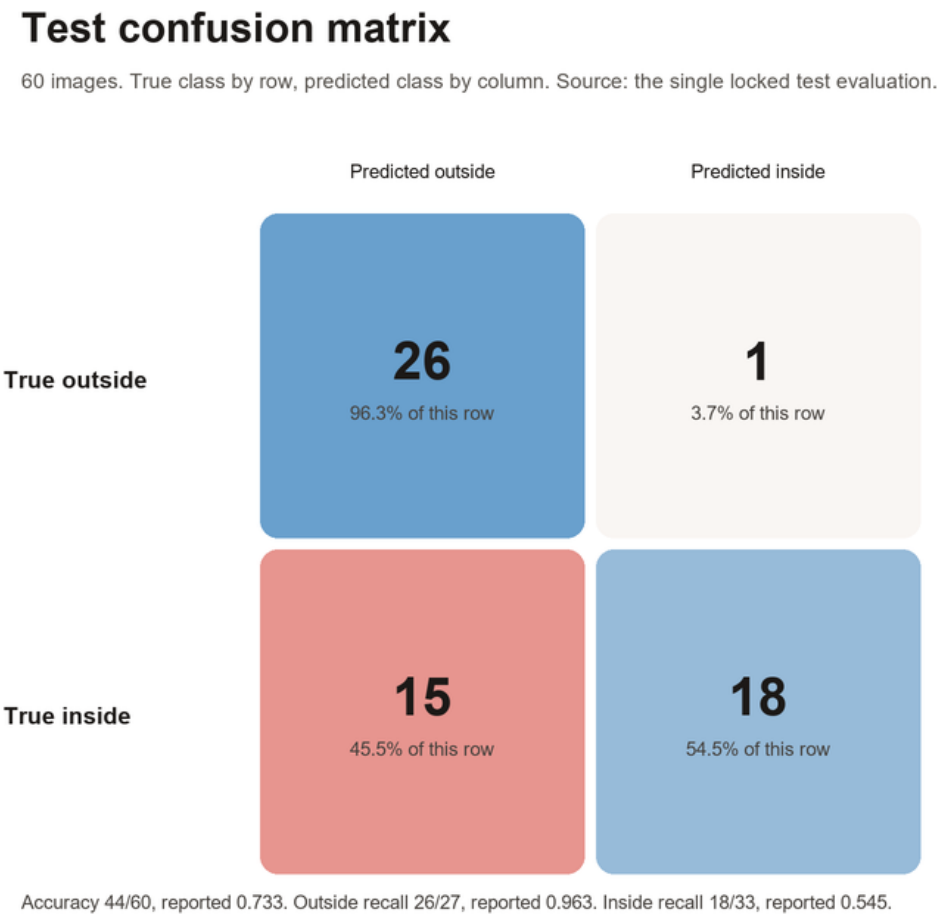


Figure 5. Test confusion matrix. Same locked pipeline, evaluated once.

Two constant classifiers frame this result. The model is ahead of both on both metrics, so neither of them is the right comparison on its own, and what they show is the size of the margin rather than a tie.

Predictor	accuracy	balanced accuracy	outside recall	inside recall
Always outside	$27/60 = 0.450$	0.500	1.000	0.000
Always inside	$33/60 = 0.550$	0.500	0.000	1.000
Locked pipeline	$44/60 =$ **0.733**	**0.754**	0.963	0.545

The test split holds 33 inside images and 27 outside ones, because the split unit is the place and the two held-out inside places are larger than the two held-out outside places. Both constants are beaten on both metrics: the always-inside constant scores 0.550 on accuracy and the always-outside constant 0.450, while the model scores 0.733; on balanced accuracy the model is 0.254 above either. The margin over the stronger constant is therefore 0.183, which is not a thin margin. The model keeps 0.963 outside recall where the always-inside constant has 0.000, and when it does say inside it is right 18 times out of 19.

The two errors are not symmetric for an application. There is one false inside in the whole test set, the OUT-06 frame that shows the Hong Kong University Main Building. There are 15 missed campus photographs, and Section 5.2 shows that 12 of the 17 frames of the CMA courtyard are among them. The low inside recall is the main limitation for an application where missing campus scenes is undesirable.

One weakness of the evidence belongs in the other direction. The selection score was fixed before the run and it was balanced accuracy, and the margin between the locked configuration and the runner-up is 0.009 from the unrounded means, well inside the fold-to-fold spread that Section 4.2 records. The test set also rests on four places, one of which, IN-07, carries 17 of the 60 images and accounts for 12 of the 15 misses. A different fold design or a different pair of held-out places would move this number, and the report quotes one evaluation of one pipeline rather than a range.

---

5. Error analysis

This section reads the test matrix place by place, then the frames. **Qualitative conclusion up front:** the locked HOG forest recognises the two unseen outside places almost perfectly, but fails unevenly on campus — mainly the CMA courtyard — and the contact sheet shows misses where the crop loses the place's identifying mark (crest, open forecourt shape) and keeps brick/planting instead.

5.1 How to read the test matrix

The four cells are not one story. 26 of 27 outside images are correct and 1 is called inside (the only false inside); 15 of 33 inside images are called outside, and that is the failure. Outside precision is 0.634 because those 15 campus misses sit with the 26 true outsides; inside precision is 0.947 because the inside column is almost clean. Accuracy 0.733 hides that split. Balanced accuracy 0.754 sits above the 0.722 holdout mean: the four held-out places happen to be easier than the average held-out training place; it does not mean the pipeline is stronger than the selection score suggested.

5.2 Failure by place

Group	Correct / n	Accuracy	Where the errors go
OUT-06 HKU Main Building	19/20	0.950	1 image predicted inside. This is the only false inside in the matrix.
OUT-07 HKU campus green	7/7	1.000	None.
IN-05 Main entrance forecourt	13/16	0.812	3 images predicted outside.
IN-07 CMA courtyard	5/17	0.294	12 images predicted outside.

Check: 19+7+13+5 = 44 (numerator of 44/60). Figure 6 plots these four accuracies against overall test accuracy 0.733.

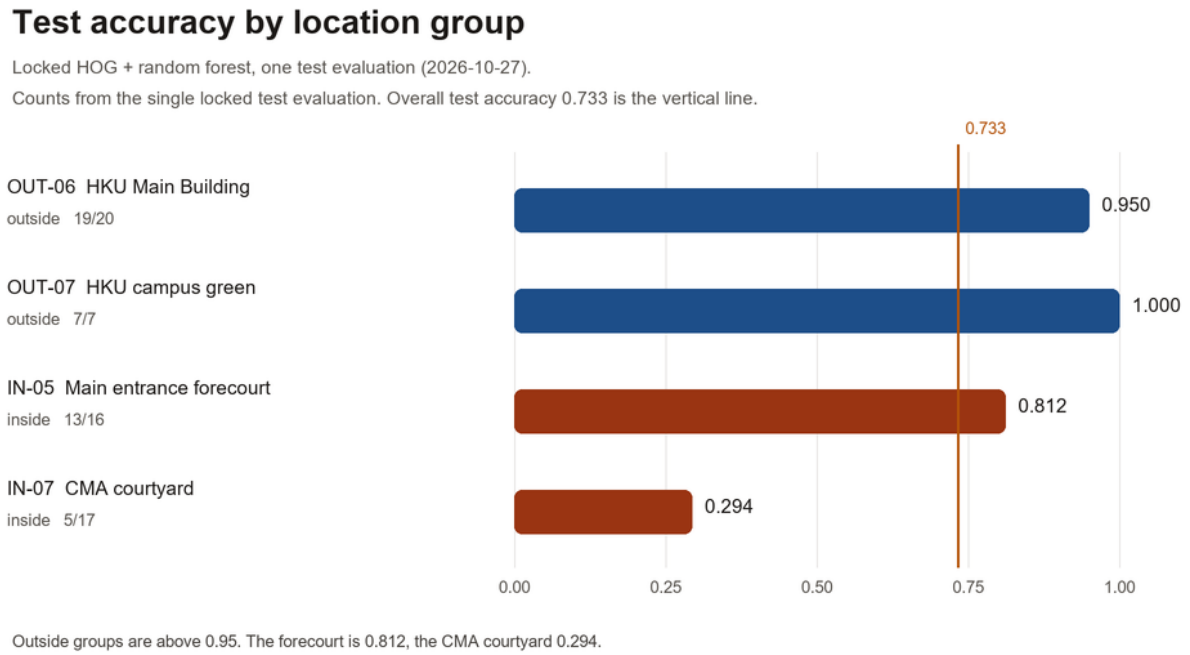


Figure 6. Test accuracy of the four held-out places. The vertical line is the overall test accuracy 0.733.

Five points: (1) the two unseen campus places are not equally hard — the courtyard alone carries 12 of 15 misses; (2) outside places are almost all correct; (3) the red-brick hard negative is rejected as PolyU on 19/20; (4) the forecourt is the easiest campus place and also the one whose frames most often fill the crop with the red-brick gate building; (5) the green pair does not transfer (OUT-07 7/7 outside; IN-07 5/17). Inside recall is 0.775 on validation and 0.545 on test; selection mean inside recall on F2+forest was 0.622 — same direction, different fits. On the held-out campus places the model misses 15 of 33 campus images.

5.3 Relation to HOG

Adding LBP to HOG does not help here: under linear SVM +0.007, under forest −0.009 (unrounded), both inside fold spread. The locked feature is HOG alone. HOG counts edge orientation in 16×16 cells; it does not encode which building or which side of the campus boundary. Training campus places (J–VA, P/Q, Library/E, FSCH/pool) need not match test places (forecourt, CMA courtyard) in those edge statistics.

Colour scored 0.537 in Stage A (fifth of six) and was not selected — consistent with the design that brick must not be a free cue. Bag of words scored 0.531; spatial pyramid 0.544. A mechanism consistent with the feature definition is that a new campus layout can fall where the forest has mostly seen outside training places.

Candidate explanation	What is measured here
Similar-looking places	Measured. OUT-06 vs forecourt (brick), OUT-07 vs IN-07 (green). Outside recognised; forecourt 0.812, courtyard 0.294.

Candidate explanation	What is measured here
Background bias	11 of 13 scene types in both classes. Errors cluster strongly by location. The current analysis does not separate location-specific effects from background bias.
Lighting	Night shares close (10.2% / 13.6% usable). Test night frames are all IN-05 (best inside place); IN-07 has none. Night does not explain the courtyard misses.
Viewpoint	No viewpoint column in the prediction table. Section 5.4: what the crop contains separates correct from missed more clearly than distance.
Road-frontage imbalance	10 vs 57 in the usable pool; test places are not that road set.

This is the score of this HOG pipeline on unseen places under a protocol that refuses to train on them — one configuration, one seed, one fold design.

5.4 Qualitative examples

Figure 7 follows this discussion. The sheet uses locked per-image test predictions and photographs from `data/pictures/` (the archive named in `DATASET_LINK.txt`). Four quadrants match the matrix cells (26, 1, 15, 18). Thirteen frames were chosen by a rule fixed in advance (filename order within each cell), recorded in `docs/figures/task4_example_manifest.csv`.

The one false inside is `OUT-06_012_IMG_0892.jpg` (glass entrance/steps at HKU, not the red-brick facade). Missed insides include close views of the red-brick name-wall at IN-05 and brick/planting at IN-07. Correct forecourt frames are wider plaza views; correct CMA frames keep the PolyU crest or a white canopy. **What the sheet supports:** misses are frames where the identifying mark is absent or replaced; the false inside is a glass entrance that looks campus-like. **What it does not support:** brick as such as the cause (a correct courtyard frame is brick too). Viewpoint and collection coverage are confounded with material here; the prediction table has no viewpoint column (Section 5.3).

5.5 The test numbers do not retune the model

Accuracy 0.733 and inside recall 0.545 are the score of the pipeline locked on location-holdout balanced accuracy. IN-05 and IN-07 stay in the test set. Features are not re-ranked on these 60 images. A later check needs new places and a new held-out evaluation of this same pipeline.

---

6. Application

The local program lets a user select one scene photograph and returns "Inside PolyU main campus" or "Outside PolyU main campus". It is a desktop window written with Tk from the Python standard library; nothing was added to `requirements.txt`. The application loads the saved pipeline and applies the same preprocessing as evaluation (Section 3.2: 256×256 centre crop). The classifier is the locked HOG random forest in `models/task3_pipeline.joblib`. Launch lines are also in the root `README.md`.

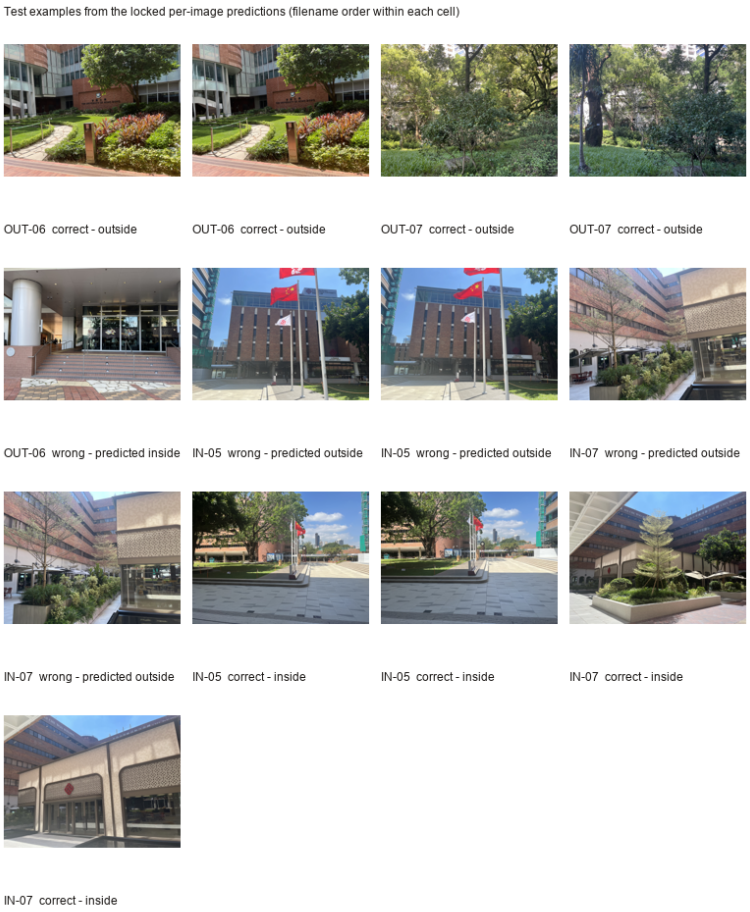


Figure 7. Thirteen test photographs in the order of the manifest: each caption gives the location group and whether the locked pipeline classified that frame correctly. The file name of every frame is in `docs/figures/task4_example_manifest.csv`.

6.1 How to launch

From the repository root, in the same environment as the training code:

```
python -m src.task5.app
```

With no display, print one image without a window:

```
python -m src.task5.app --image path\to\photo.jpg
```

On Windows, `pythonw -m src.task5.app` opens the window without a console. If `models/task3_pipeline.joblib` is absent, the window tells the user to run `python -m src.task3.evaluate`; in this submission the file is present.

6.2 What the window shows

Select image... (or Ctrl+O) opens a file dialog. The left pane shows the photograph; the right pane prints the class sentence and, for this forest, both class probabilities; the thumbnail is the 256×256 crop the model receives. A bottom line names the pipeline file and states that file name, folder, EXIF and GPS are not used. The application loads the saved pipeline and applies the same preprocessing as evaluation.

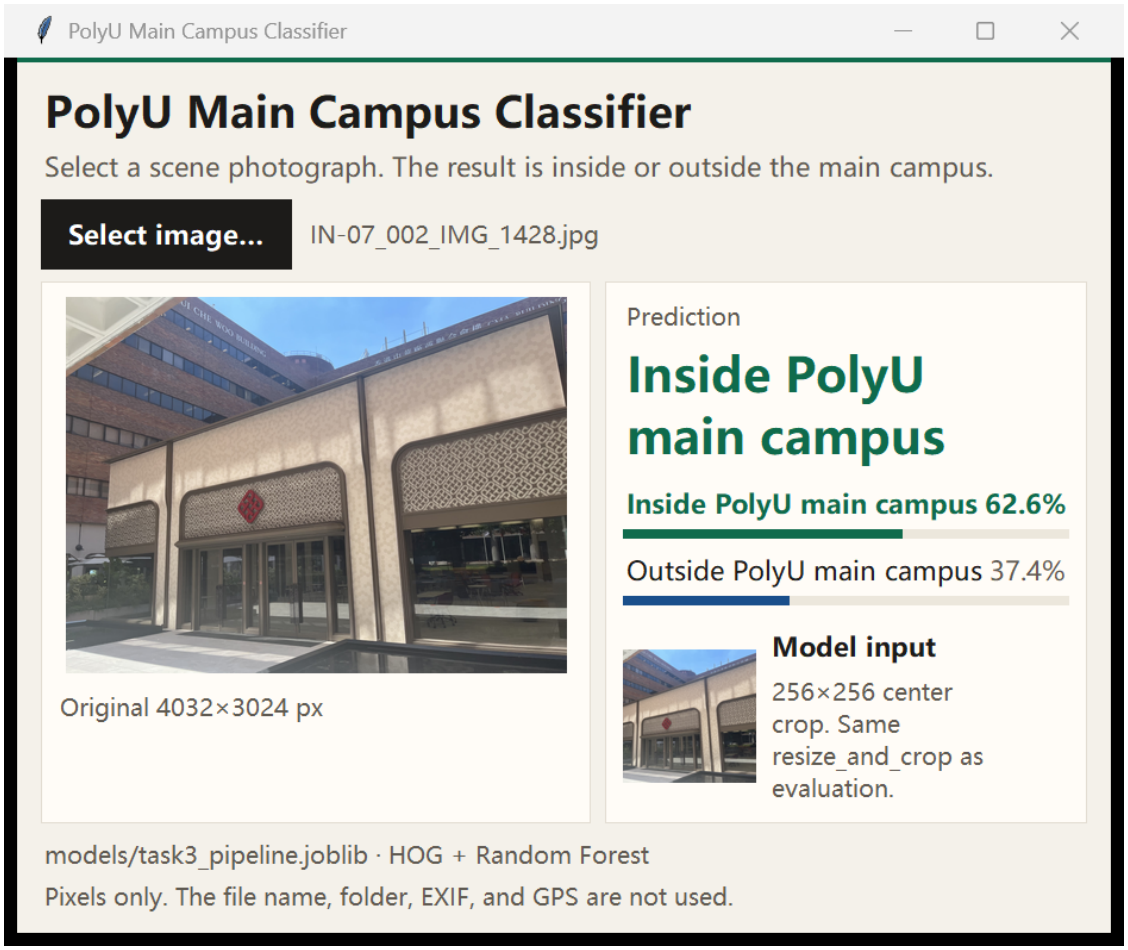


Figure 8. The application with a campus photograph loaded (`IN-07_002_IMG_1428.jpg`, CMA courtyard with the PolyU crest). The locked pipeline calls it inside with 62.6%.

6.3 Checks and real runs

Checks on the installed `requirements.txt` (no refit): mean 0.68 s per image over 20 test frames; crop shape always `(256, 256, 3)`; same pixels under `IN-99_...` and `OUT-99_...` names give the same label and probabilities; unreadable file keeps the previous label / exits 1 on `--image`; all 60 test images replayed through the application prediction path matched the locked prediction table 60/60. Two course-material halves (Figure 1 of the assignment) both classified correctly with probabilities within 0.013 of 0.5 — successful inference, not a generalisation claim. Held-out test remains accuracy 0.733, inside recall 0.545 (Section 5).

```
python -m src.task5.app --image \
  data/pictures/inside/U_Garden_Area/IN-07_002_IMG_1428.jpg
Inside PolyU main campus
outside: 0.374
inside: 0.626
```

```
python -m src.task5.app --image \
  data/pictures/outside/HKU/OUT-06_001_IMG_0863.jpg
Outside PolyU main campus
outside: 0.717
inside: 0.283
```

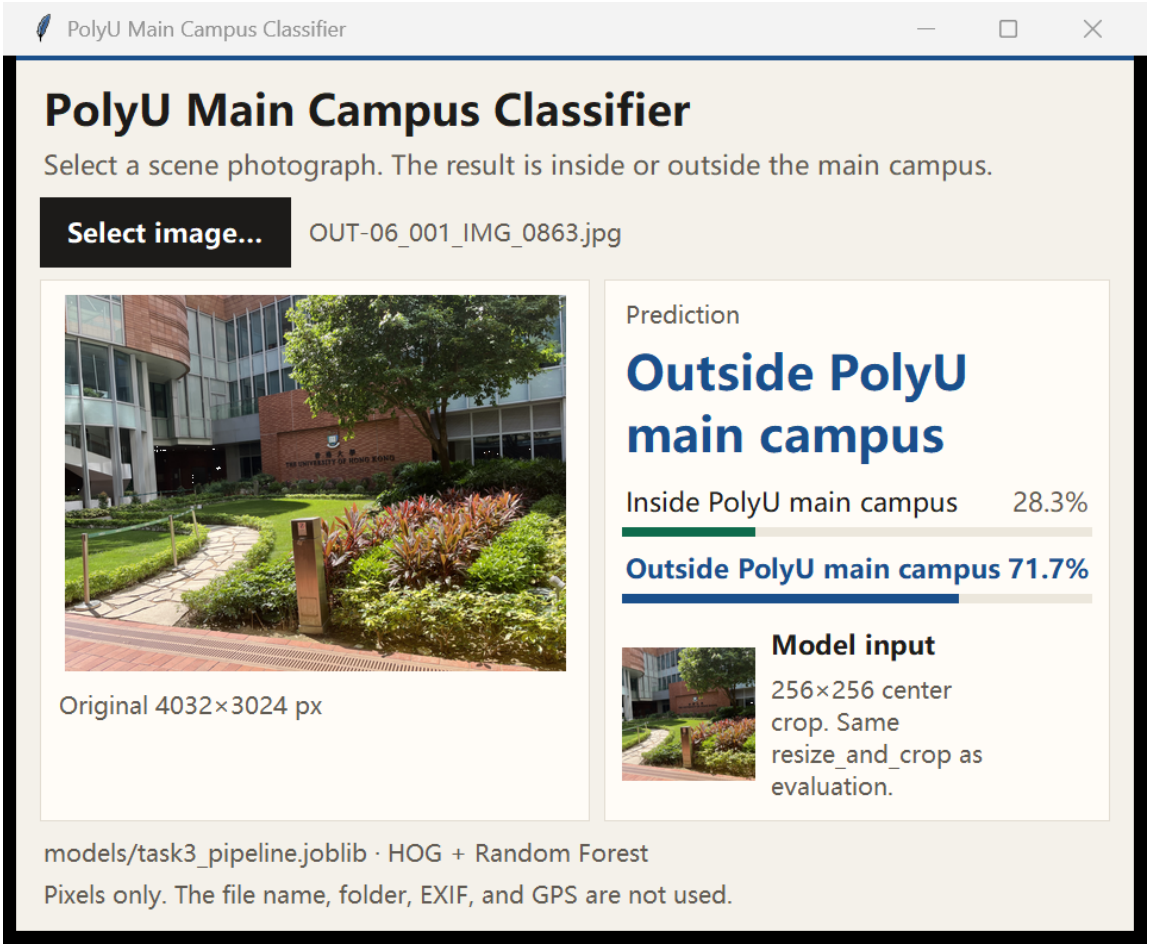


Figure 9. The same application with an outside photograph (`OUT-06_001_IMG_0863.jpg`, HKU Main Building). The locked pipeline calls it outside with 71.7%.

Figures 8 and 9 are screenshots of that same prediction path. Test accuracy figures in Sections 4–5 are unchanged by the application.

---

7. Discussion

Six template questions, answered briefly with cross-references rather than restating the pipeline.

How do you design and test the program?

Geographic rules first (campus vs not, boundary frames dropped, place as split unit), then matched scene types and hard negatives (Section 2). Learning is two locked stages: Stage A six features × linear SVM, Stage B classifiers on F4/F2, score = mean location-holdout BA (Section 3.1). Automatic split checks and published tables match stored Stage A/B and test predictions (Section 3.6). The desktop program is checked separately (Section 6.3): 60/60 replay, name-invariant pixels, second crop is a no-op.

How do you ensure the robustness of the program in a real scenario?

Three claims, supported differently. **Pipeline consistency** — measured: evaluation and the window share the same crop path and `models/task3_pipeline.joblib`; 60/60 replay (Section 6.3). **Input handling** — bad files do not crash the window; crop cannot be applied twice by accident. **Predictive robustness on unseen places** — measured and weak: test inside re-

call 0.545, CMA courtyard 5/17 (Section 5). Lighting and portrait confidence are thin on the evidence available. Raising the third claim needs more unseen campus places under the same rules, not moving current test groups into training.

What problems do you find and how do you solve them?

Problem	What was done
GPS lost after copy	Labels from signs in frame (Section 2.3); GPS never a feature.
Ambiguous gate / people / plates	Excluded with manifest notes.
Red brick as inside shortcut	HKU brick collected as outside; colour histogram fifth of six (0.537).
Easy validation pair / validation vs holdout disagree on second feature	Validation reported only; holdout ranks (Section 4).
Folder-based grouping errors (leakage)	Photo-level recut, 31 moves / 2 drops; rebuild (Section 2.7).
Test inside recall 0.545	Breakdown in Section 5; no retune on test; remedy is new held-out places (Section 8).
Class-like file names	Prediction uses pixel arrays only; file names are not features (Section 6.3).

How do you use GenAI to assist you in the implementation?

AI drafted feature/classifier/holdout/dataset/figure code under a protocol I fixed first (five folds, one selection score, test once on the repaired split). I decided place labels, include/exclude, and any disagreement between a generated note and the photograph. Verification: recompute Stage A/B and the confusion matrix from stored tables; re-read the four repaired groups by eye (Sections 2.7, 4.2, 6.3).

How does GenAI understand and solve the tasks?

It handles the classical pipeline recipe well (parse split, crop, feature, fit, write metrics). It is weaker on facts that live only in photographs or campus-specific rules:

- The campus-map annotation was first generated from a coarser place list with ids that appear nowhere in `splits.csv`; the figure was rebuilt from the split file.
- The CMA building was first recorded as 香港中華煤氣. The sign in the photograph reads 香港中華廠商聯合會樓 (CMA Building). The manifest entry was corrected from the sign.
- License-plate characters taken from an annotation rather than the image were wrong; the frame is excluded on privacy grounds regardless.
- Source manifests were left out of version control by a generated ignore rule; the rule was changed and the manifests committed.
- When SIFT might be missing, the suggestion was to switch to ORB; the rule in Section 3.3 is to stop. The run kept SIFT because `cv2.SIFT_create` was present.

The holdout column — not validation — ranked features (Section 4.1–4.2).

What are the limitations and areas for improvement in GenAI's solution?

Treat image-content and place-identity claims as unchecked until verified against the file; hard-stop if SIFT is missing; freeze the selection column before the run; check location groups against

photographs (Section 2.7). The model-side limit in Section 5 is separate and is not repaired by generating against the test score.

---

8. Reflection

What I expected from a protocol fixed in advance was a ranking that would hold up, and a score that would carry to new places. The first expectation was only partly right, and the second was wrong.

The protocol did what it was built to do at the level of the rules: the colour shortcut was rejected, the codebook was rejected, and no classifier was chosen by validation accuracy. What the rules did not control was how much the score moves between folds. With one seed and effectively four independent inside training places, the fold-to-fold spread is wider than the gaps between the top configurations, so the order of the top rows is not something this much data can settle.

What the result taught is that the score carries less than the design assumed. The selected pair, HOG with a random forest at 0.722 location-holdout balanced accuracy, becomes accuracy 0.733 and inside recall 0.545 on the held-out test places, with OUT-06 at 0.950, OUT-07 at 1.000, IN-05 at 0.812, and IN-07 at 0.294. The outside places transferred almost perfectly and one campus place did not. The holdout had suggested a weaker version of the same asymmetry, with inside recall 0.622 against outside recall 0.821, but nothing in it suggested a drop to 0.294 on the hardest of the four places.

Those test numbers are a usable measurement and a poor campus detector. The representation has no separate channel for "this is PolyU" once colour is set aside and the building has not been seen. What transfers is local gradient, and the failure is consistent with a façade that resembles nothing in the training places, which is what the CMA courtyard is; that reading is a hypothesis, not a measurement. Whether the limit is the material or the layout is not settled here, and the test set was not refit to find out. A second lesson came from the repair in Section 2.7: a location group is a claim about photographs, and a claim made from a folder name was wrong for four of the sixteen, so the groups had to be re-read and the whole pipeline rebuilt before any of the numbers above could be called a measurement.

What would actually change the evidence is more inside places, collected under the same boundary rules, and one new held-out evaluation of this saved pipeline. The route is concrete: add inside places so that a fifth inside group can be held out, leave the current IN-05 and IN-07 groups in the test set and never train on them, and score the new group with the pipeline saved in `models/task3_pipeline.joblib` rather than with a refit.

Two further measurements are worth making, and they cost the current result very different amounts.

The cheap one is a per-cell breakdown that also splits by scene class. It reads the 60 predictions that already exist, adds no fit, and does not touch the locked model; the aggregate table cannot show which part of a frame the model actually saw.

The expensive one is a feature that keeps a cue the locked one throws away. HOG is computed on the grayscale crop, so colour is not available to it, and a variant that adds a colour histogram would separate "the crop was mostly brick and paving" from "the negative was never learnable" (Section 5.4 shows the misses clustering on crops where the identifying mark is missing or replaced by something adjacent). **That variant is a new pipeline, not a reading of the cur-**

rent one. It needs its own fit, its own location-holdout selection and its own held-out evaluation, and it would replace the saved model rather than extend it. `models/task3_pipeline.joblib` stays the baseline the candidate is measured against, and the 0.733 in Section 4.4 is the baseline's number, not the candidate's.

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9. References

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10. Figure index

Figures 1–9 are embedded above; files live in `docs/figures/` in the code archive (pipeline, maps, confusion matrices, per-group bar, example grid, two application screenshots).

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11. Appendix: terms and abbreviations

Feature ids F1–F6 and classifier ids C1–C6 are defined in the tables of Sections 3.3 and 3.4. The locked pair is **F2 HOG + C4 random forest**. Standard evaluation terms (balanced accuracy, location holdout, hard negative, precision/recall) are used in their usual senses; the selection score is mean location-holdout balanced accuracy (Section 3.5). Split / group_id / seed (42) are as in Sections 2.5 and 3.6.

Location-group repair ledger (Section 2.7)

Sizes the split rule read after the 2026-10-27 photo-level repair (groups whose split or stored count changed):

Group	Split before	Split after	Frames before	Frames after
IN-03	test	train	25	44
IN-05	validation	test	38	16
IN-06	train	validation	40	40
OUT-04	train	train	30	29
OUT-08	train	train	30	29
OUT-09	validation	validation	30	29

The other ten groups kept both split and stored count. IN-03 gained 19 (13 from IN-05, 6 from IN-01); IN-05 lost 22; IN-01 stays capped at 50. The two exclusions came from a capped group, so they cost no kept frame; the lower inside total after moves is what the balance step re-reads (target 267, outside share 29 rather than 30).